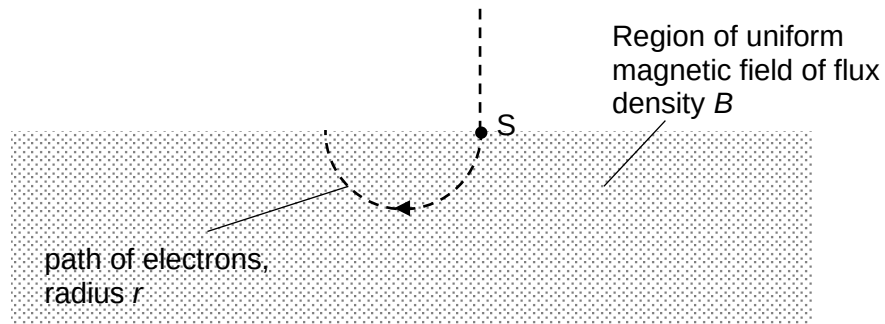


- 24 Electrons, each of mass m and charge q , are accelerated from rest in a vacuum through a potential difference V .

The accelerated electrons are then projected at point S into a region of uniform magnetic field of flux density B , as shown. The electrons move in a circular path of radius r .



Which of the following expressions represents the specific charge $\frac{q}{m}$ of the electrons?

- A $\frac{V}{2B^2r}$ B $\frac{2V}{B^2r}$ C $\frac{V}{2B^2r^2}$ D $\frac{2V}{B^2r^2}$

Ans: D

$$qV = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2qV}{m}}$$

$$Bqv = \frac{mv^2}{r}$$

$$Bq\sqrt{\frac{2qV}{m}} = \frac{m\left(\frac{2qV}{m}\right)}{r}$$

$$\sqrt{\frac{2qV}{m}} = \frac{m\left(\frac{2V}{m}\right)}{Br} = \frac{2V}{Br}$$

$$\frac{2qV}{m} = \frac{4V^2}{B^2r^2}$$

$$\frac{q}{m} = \frac{2V}{B^2r^2}$$

25. A rectangular coil made of 100 turns of wire with cross sectional area 20 cm^2 is placed within a